

MS programs in Statistics and Biostatistics at American University

College of Arts & Sciences > **Mathematics & Statistics**



Department of **Math & Statistics**

Offering graduate and undergraduate programs in Biostatistics, Data Science, Mathematics, Quantitative Analysis, and Statistics.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

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<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

HIGHLIGHTS

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

Study Statistics in the [USA Capital!](#)

American University in Washington DC offers the MS degree in [Statistics](#), featuring:

- ❑ Rigorous training in statistical theory, methods, and applications;
- ❑ Small classes, low students to faculty ratio;
- ❑ Individual attention and individualized degree plans according to your career goals;
- ❑ Practical experience: research, teaching, consulting, and internship opportunities under faculty guidance;
- ❑ Two tracks: *Theory and Methods* and *Biostatistics*;
- ❑ Optional completion of a Graduate Certificate in *Data Science* while you pursue the MS degree, making you a formally certified specialist;
- ❑ *Financial support*: partial tuition remission, teaching and research assistantships on merit basis;
- ❑ Incoming M.S. in Statistics students are expected to have completed Calculus I–III and intermediate-level Statistics. However, we can tailor a degree plan for students who may be missing a prerequisite.

Related Links:

- MS program in Statistics (including Biostatistics)
 - www.american.edu/cas/mathstat/statistics-ms/
- Math & Stat Department
 - www.american.edu/cas/mathstat/
- Course descriptions
 - <https://catalog.american.edu/>, go to Courses
- Our professors and their research
 - <https://www.american.edu/cas/mathstat/faculty/index.cfm>
- Course schedules
 - <http://www.american.edu/provost/registrar/schedule/index.cfm>
- International Students: here you can ask any related questions
 - <https://www.american.edu/student-affairs/iss/>
- Events and campus visits: schedule in-person and virtual visits here
 - <https://www.american.edu/cas/admissions/events.cfm>
- Submit application
 - <https://www.american.edu/cas/admissions/apply.cfm>

CONTACT INFORMATION

AMERICAN UNIVERSITY MATHEMATICS & STATISTICS DEPARTMENT

<https://www.american.edu/cas/mathstat/statistics-ms/>

PROGRAM DIRECTOR, MS in STATISTICS

Dr. Michael Baron, baron@american.edu

WEEKLY Ask-Me-Anything ZOOM SESSIONS ABOUT MS in STATISTICS:

<https://american.zoom.us/j/95868480371> - every Monday at 12 noon

SCHEDULE A VISIT – in person or virtual tour

<https://www.american.edu/cas/admissions/events.cfm>

APPLY to the PROGRAM:

<https://www.american.edu/cas/admissions/apply.cfm>

Ask-Me-Anything Weekly Zoom Sessions

Every Monday at 12:00 – 1:00 pm ET

Link: <https://american.zoom.us/j/95868480371>

Connect, meet the MS-Statistics program director and ask about:

- Unique opportunities at American University
- Careers in Statistics
- Financial packages
- How to apply
- Curriculum, required and elective courses
- Research opportunities
- Internship opportunities
- International students
- Student life, student clubs
- Etc, etc, etc.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: Dr. Michael Baron - baron@american.edu

OUR PROGRAMS

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

MASTERS DEGREE IN STATISTICS

STATISTICS (MS)

This program prepares professionals in the development, research, and analysis of quantitative tools to turn ideas into intelligent, informed actions.

PROGRAM TRACKS

The [Statistics MS](#) offers two tracks of study: Theory and Methods and Biostatistics. Each of them can be combined with the Graduate Certificate in Data Science.

THEORY and METHODS

Rigorous training in Statistics that is expected for M.S. level jobs and applications for Ph.D. programs.

BIOSTATISTICS

Solid training in Statistics with Biostatistics and Biology components.

ADMISSION TO THE PROGRAM

Applicants are expected to have a bachelor's degree from an accredited institution with a cumulative grade point average of 3.00 on a 4.00 scale and an adequate background in the mathematical sciences*.

*Prerequisites: Calculus I-III, matrices, and an intermediate level course in Statistics.

DEGREE REQUIREMENTS

30 credit hours of graduate coursework including

- Mathematical Statistics
- Regression
- Statistical Machine Learning
- Advanced Machine Learning courses
- Deep Learning and Neural Networks
- Statistical Software, R and/or SAS
- Generalized Linear Models
- Research / Consulting / Internship experience

Up to 9 approved graduate credit hours may be shared between the BS and MS degrees in a [combined BS-MS program](#). Ask for information.



EMPLOYMENT OPPORTUNITIES

Departmental ties to governmental agencies such as the National Science Foundation, the Environmental Protection Agency, the Food and Drug Administration, and the National Institutes of Health offer students exposure to numerous research and employment opportunities. Currently, 100% of recent MS graduates in Statistics from AU are employed or pursuing their next degree (see www.american.edu/weknowsuccess/).

CONTACT

AU MATH & STAT DEPARTMENT
www.american.edu/cas/mathstat/

STATISTICS PROGRAM DIRECTOR
Dr. Michael Baron, baron@american.edu

Contact any time for more information ☺



MASTERS DEGREE IN STATISTICS

with the Graduate Certificate in Data Science

Rigorous training in Statistics + practical courses in Data Science = Exceptional Employment Opportunities!

- Pursue the [MS degree in Statistics](#) and the [Graduate Certificate](#) in Data Science at the same time.
- Enter the job market as the MS degree holder in Statistics and a certified specialist in Data Science.
- Total number of required graduate courses = 10. With proper planning, the Certificate courses are also counted toward the MS degree.

STATISTICS (MS) with Graduate Certification

This program prepares professionals in the development, research, and analysis of quantitative tools to turn ideas into intelligent, informed actions.

PROGRAM TRACKS

The [Statistics MS](#) offers two tracks of study: Theory and Methods and Biostatistics. Each of them can be combined with the Graduate Certificate in Data Science.

THEORY and METHODS

Rigorous training in Statistics that is expected for M.S. level jobs and applications for Ph.D. programs.

BIostatISTICS

Solid training in Statistics with Biostatistics and Biology components.

ADMISSION TO THE PROGRAM

Applicants are expected to have a bachelor's degree from an accredited institution with a cumulative grade point average of 3.00 on a 4.00 scale and an adequate background in the mathematical sciences*.

*Prerequisites: Calculus I-III, matrices, and an intermediate level course in Statistics.

DEGREE REQUIREMENTS

30 credit hours of graduate coursework including

- Mathematical Statistics
- Regression
- Statistical Machine Learning
- Statistical Software, R and/or SAS
- Data Science
- Advanced Machine Learning courses
- Deep Learning and Neural Networks
- Research / Consulting / Internship experience

Up to 9 approved graduate credit hours may be shared between the BS and MS degrees in a [combined BS-MS program](#). Ask for information.

EMPLOYMENT OPPORTUNITIES

Departmental ties to governmental agencies such as the National Science Foundation, the Environmental Protection Agency, the Food and Drug Administration, and the National Institutes of Health offer students exposure to numerous research and employment opportunities. Currently, 100% of recent MS graduates in Statistics from AU are employed or pursuing their next degree (see www.american.edu/weknowsuccess/).

CONTACT

AU MATH & STAT DEPARTMENT
www.american.edu/cas/mathstat/

STATISTICS PROGRAM DIRECTOR
Dr. Michael Baron, baron@american.edu

Contact any time for more information ☺



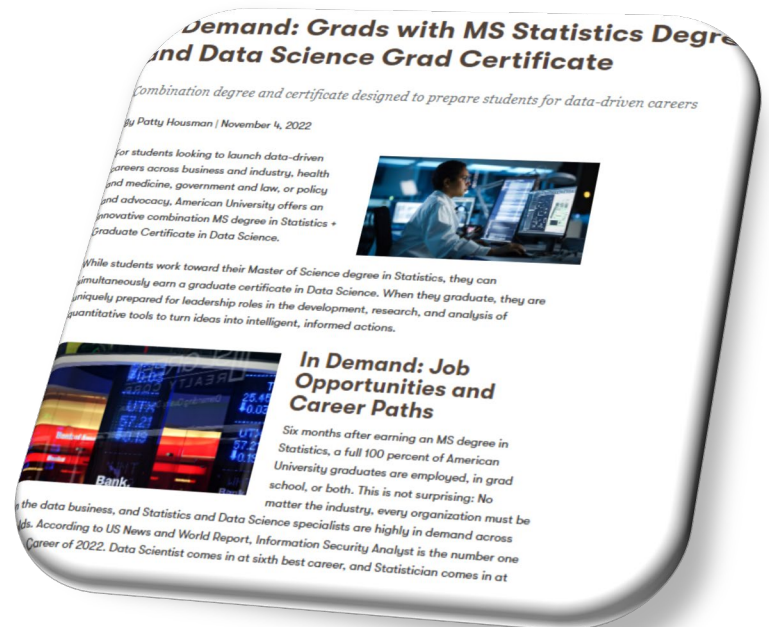
MS DEGREE COMBINED WITH A GRADUATE CERTIFICATE

No additional cost or additional courses,
with proper planning

Click to read [an article](#) about
this **MS + Certificate** program

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



Masters Degree in Statistics with Graduate Certificate in Data Science

*American University
Washington DC*

1 Customized degree plans are tailored to meet your goals

2 No extra courses: Certificate is integrated within the MS program

3 Financial support packages are available

Two tracks:

- Theory and Methods
- Biostatistics

Statistics + Data Science
30 graduate credit hours

✓ Rigorous training in Statistics



✓ Practical courses in Data Science and Machine Learning



✓ *Exceptional
Employment
Opportunities*

Graduate program in the Nation's Capital



Research and internship opportunities

Active student clubs: networking, workshops, community events



CONTACT for info and join our zoom sessions:

Michael Baron,
MS program director
baron@american.edu



ACCELERATED BS-MS PROGRAM

Relevant to the current AU students

How it works:

- Start taking relevant graduate courses during your undergraduate program at AU
- Share up to 9 credit hours between your BS and MS programs
- Graduate with MS degree in Statistics 1 year after receiving the BS degree

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*



DOUBLE-EAGLE SCHOLARSHIP

Guaranteed 30% tuition discount for your MS program in Statistics.

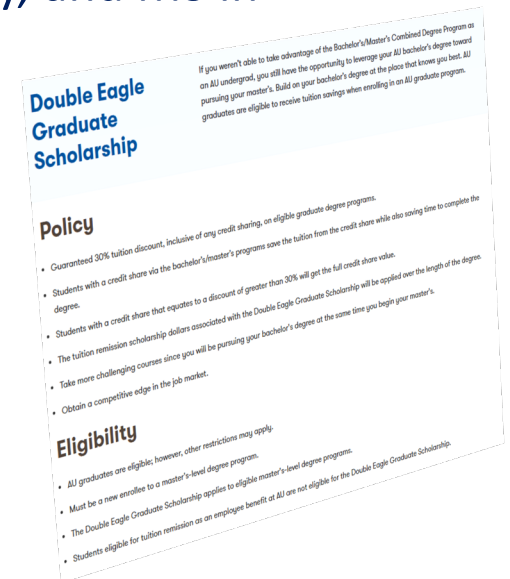
You are eligible if:

- You already have a degree from American University, and MS in Statistics is your second degree.
- You are not in an accelerated BS-MS program.

Details: <https://www.american.edu/provost/ogps/graduate-studies/double-eagle-graduate-scholarship.cfm>

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



SAMPLE DEGREE PLANS

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

MS in Statistics (Statistical Science track)

Course		Semester	Grade	Prerequisites
<i>Required core (12 hours):</i>				
STAT 615	Regression			Stat 614 Statistical Methods or equivalent
STAT 627	Statistical Machine Learning			Stat 615 Regression
STAT 630	Mathematical Statistics I			Three semesters of Calculus, generally
STAT 631	Mathematical Statistics II			Stat 630 Math Stat I
<i>Software (choose one, 3 hours):</i>				
DATA 612	Statistical Programming in R			or equivalent
STAT 625	Statistical Software (SAS)			Stat 614 Statistical Methods or equivalent
<i>Data collection (choose one, 3 hours):</i>				
STAT 516	Design of Experiments			Stat 614 Statistical Methods or equivalent
STAT 605	Intro to Survey Sampling			Stat 614 Statistical Methods or equivalent
<i>Capstone options (6 hours, thesis or nonthesis options)</i>				
STAT 690	Independent Project			Department permission, register via a signed form
STAT 691	Internship in Statistics			Department permission, register via a signed form
STAT 797	Master's Thesis Research			Department permission, register via a signed form
STAT 798	Research & Consulting			Department permission, register via a signed form
<i>Other electives (500 or 600 level courses in STAT or DATA, 6 hours)</i>				
STAT xxx				
STAT xxx				
DATA xxx				

Total credit hours = 30 (~ 10 graduate courses)

Typical duration = 16 to 21 months. Part-time enrollment is possible.

MS in Statistics (Biostatistics track)

Course		Semester	Grade	Prerequisites
Required core (15 hours):				
STAT 615	Regression			Stat 614 Statistical Methods
STAT 622	Advanced Biostatistics			Stat 614 Statistical Methods
STAT 623	Topics in Biostatistics			Repeatable for credit, on a different topic
STAT 630	Mathematical Statistics I			Three semesters of Calculus, generally
STAT 631	Mathematical Statistics II			Stat 630 Math Stat I
Biology elective (3 hours)				
BIO 68x	BIO-685 Bioinformatics, or BIO-687 Principles of Genomics, or BIO-689 Biotechnology, or.			
Capstone options (6 hours , thesis or nonthesis options)				
STAT 690	Independent Project			Department permission, register via a signed form
STAT 691	Internship in Statistics			Department permission, register via a signed form
STAT 797	Master's Thesis Research			Department permission, register via a signed form
STAT 798	Research & Consulting			Department permission, register via a signed form
Choose a focus and 6 hours :				
Health Policy Focus - 6 hours from COMM 580, ENVS 665, HLTH 575, 585, 661, 670, PUAD 604, 685, 696, SIS 624, 626, 628, 635				
Computational Focus - 6 hours from CSC 589, 610; ITEC 616, 620, 660, 670; MATH 660, 665; SIS 646; STAT 612, 625, 627				
Management Focus - 6 hours from ACCT-600, COMM-543, FIN-630, ITEC-616, ITEC-620, ITEC-660, MKTG-612, PUAD-685, SIS-628, SIS-635, SIS-646				
Or choose electives (500 or 600 level courses in STAT or DATA, 6 hours)				
STAT xxx	Choose courses in Statistics, Biology, Health Studies, Machine Learning, Artificial Intelligence, Data Science, to match your interests and career goals			
DATA xxx				

Total credit hours = 30 (~ 10 graduate courses)

Typical duration = 16 to 21 months. Part-time enrollment is possible.

Graduate Certificate in Data Science

Course		Prerequisites
<i>Each following course is required (12 credit hours)</i>		
STAT 615	Regression	Stat 614 Statistical Methods
DATA 612	Statistical Programming in R	Stat 614 or two Math/Stat courses 200+ level
DATA 613	Data Science	DATA 412/612
STAT 627	Statistical Machine Learning	STAT 415/615 or STAT 520
STAT 614	Statistical Methods	<i>Pre-requisite for the program</i>

Total courses = 4

All 4 courses can be shared between the Certificate and the MS degree

You graduate as a certified professional in Data Science

Graduate Certificate in Applied Statistics

Course		Prerequisites
<i>Each following course is required (6 credit hours)</i>		
STAT 615	Regression	Stat 614 Statistical Methods
STAT 605	Survey Sampling	Stat 614 Statistical Methods
<i>Electives (6 credit hours) - approved graduate courses in Statistics</i>		
STAT ____	Choose courses in Statistics, Biostatistics, Machine Learning, Artificial Intelligence, Data Science, to match your interests and career goals	
STAT ____		

Total courses = 4

All 4 courses can be shared between the Certificate and the MS degree

You graduate as a certified professional in Applied Statistics

OUR STUDENTS

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

STUDENT CLUBS



➤ Student Chapter of ASA

Advisor: Betty Malloy

➤ Girls Who Code

Advisor: Donna Dietz

➤ Not Math Club

Advisor: Mike Limarzi

<https://www.american.edu/cas/mathstat/student-clubs.cfm>

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AU Statistics & Data Science Club

Unlock Your Future in STEM 🔬💻🧬📊

Why Join🔗?

- Empower Your Skills: Leadership, teamwork, and decision-making🧑🌞📈.
- Expand Networks: Connect with industry professionals.
- Learn & Grow: Workshops and talks by industry experts.
- Build Your Career: Add valuable experience to your resume and build your portfolio.

About Us:

- Founded in January 2022 by AU's brightest in Mathematics and Statistics.
- The largest, most active STEM community at AU.

What We Offer:

- Expert Insights: Stay ahead with the latest trends and technologies, fostering out of your classroom.
- Hands-On Experience: Real-world projects🔧📊💻 with a tangible impact.
- AU-ASA STEM community: An opportunity that serves as a hub to connect, exchange knowledge, and collaborate with other AU students.

American Statistical Association @ American University

Join Us: Be part of an innovative community driving the future of data science, emerging technologies at AU.

Grow, Learn, Lead. Together👉🌐🎓.



<https://www.american.edu/cas/mathstat/student-clubs.cfm>

<https://www.american.edu/cas/mathstat/statistics-ms>

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A guide to interview preparation workshop



Internship Event



Data Visualization workshop



AU-ASA at Club Involvement fair

<https://www.american.edu/cas/mathstat/student-clubs.cfm>

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*



WHO IS GWC?

AU Girls Who Code is a chapter of the national GWC movement, which aims to empower individuals to pursue careers in tech and provide a community for like minded individuals who are excited to explore where coding skills can be used in their desired fields!

Our chapter is a community of women, gender non-conforming, and all underrepresented individuals who are interested in empowering each other and closing the gender gap in tech.

OUR MISSION

We are currently working towards building a GWC chapter at AU in order to foster a safe, supportive community for those interested in coding; whether it be in the field of computer science or a general interest!

Do you have coding experience?
Are you interested in learning to code or exploring how coding can enhance your career? Join GWC to find your community at AU! No experience necessary!

MAILING LIST

Scan this QR code for information on upcoming events, meetings, and more!



connect with us!

instagram



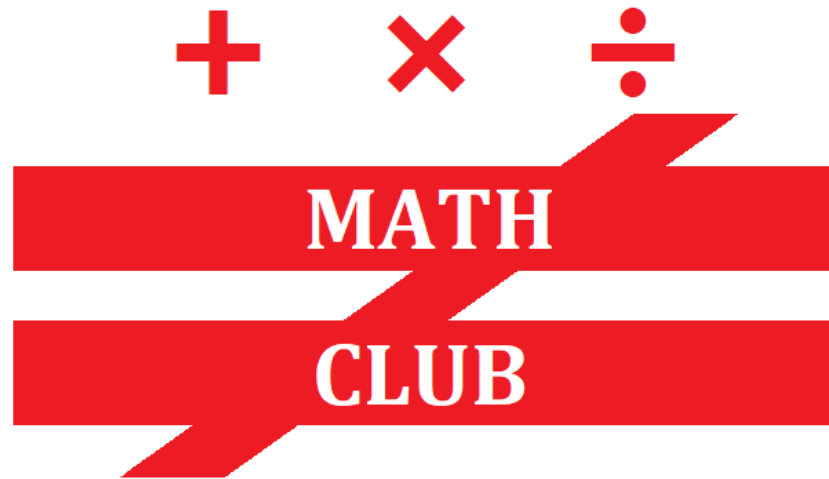
discord



Feel free to reach out to us at augirlswhocode@gmail.com with any questions, comments, or concerns!

Want to be a part of the coolest club at AU?

If not, then you should join

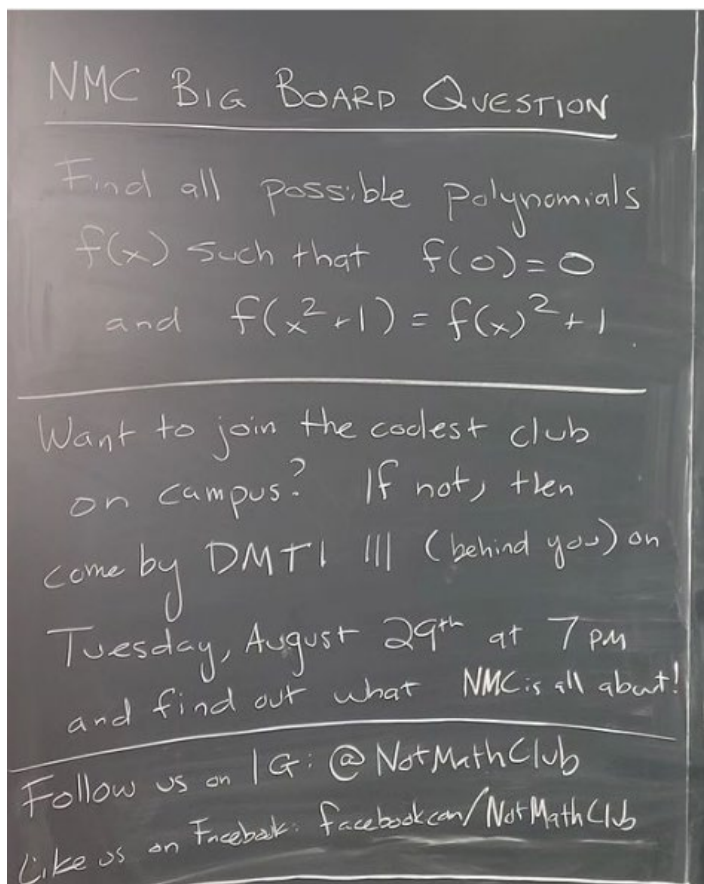


Not Math Club!

Meetings on Mondays at 7PM in DMTI 111.

We take math seriously. But not too seriously.

NOT MATH CLUB



$+$ $\times$ $\div$



Meets: Mondays @ 7 pm in
room DMTI-111

Faculty advisor: Mike Limarzi

<https://www.american.edu/cas/mathstat/student-clubs.cfm>

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

RESEARCH OPPORTUNITIES



Selected Summer Project Examples



Eyerusalem Abebe

"Analysis of graphical models using sheaves"

Supervisor: Michael Robinson



Jacqueline Adams

"Macrophages and Iron Phenotypes in Cancer"

Supervisor: Julia Chifman



Casey Aguilar-Gervase

"Gene interactions, order perturbations and algorithms"

Supervisor: Kristina Crona

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

RESEARCH OPPORTUNITIES



Tonia Bell

"Gene interactions, rank orders and
hyperplane arrangements"

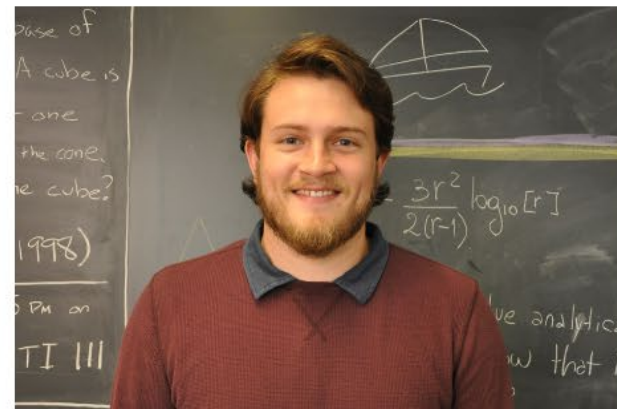
Supervisor: Kristina Crona



Payal Dudhedia

"Antibiotic resistance, order perturbations and
cube graphs"

Supervisor: Kristina Crona



Mario Ego-Aguirre

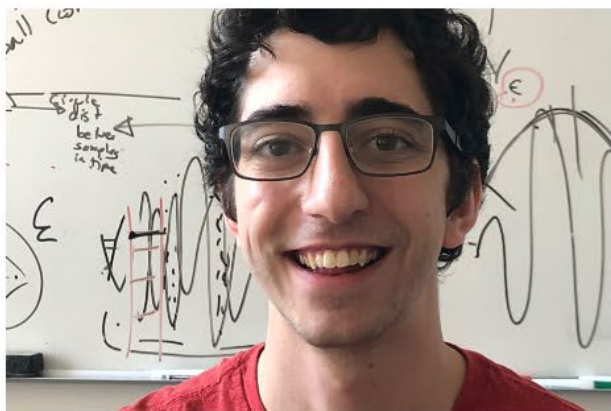
"Slices of Pi"

Supervisor: Kenneth Ward

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

RESEARCH OPPORTUNITIES



Robby Green

"Acoustic simulation and experiments of
percussive musical instruments"

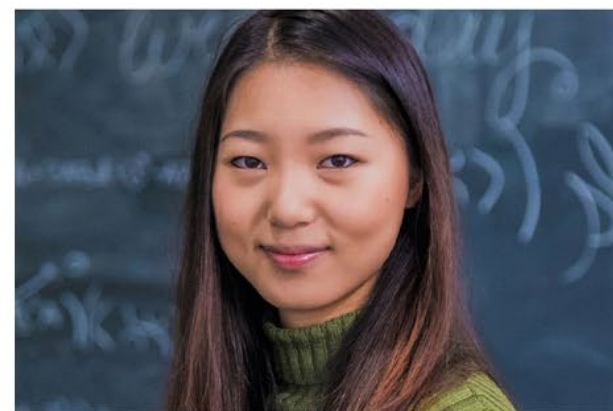
Supervisor: Michael Robinson



Joe Kelly

"Acoustic simulation and experiments of
percussive musical instruments"

Supervisor: Michael Robinson



Fangfei Lan

"Acoustic simulation and experiments of
percussive instruments"

Supervisor: Michael Robinson

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

RESEARCH OPPORTUNITIES



Moriah Mitchell

"Involvement of iron in cell cycle"

Supervisor: Julia Chifman



Metin Toksoz-Exley

"Biological Networks and Generative Graphs"

Supervisor: Michael Robinson



Greg Young

"Acoustic simulation and experiments of
percussive musical instruments"

Supervisor: Michael Robinson

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

RESEARCH OPPORTUNITIES

Meet a Researcher

Tonia Bell



Why did you want to do a research project?

I wanted to do a research project so that I could take what I learned in the classroom and see how it applies to real life situations. Seeing what you study theoretically apply in reality is a very rewarding experience that makes all the hours spent studying feel worthwhile.

What do you research?

I researched additivity in biological systems. Since any organism will adapt to better fit its surroundings, one can look at the genome and observe where mutations occur. Additive systems are those in which mutations contribute to fitness independently. By using weight vectors and hyperplane arrangements, we were able to find all rank orders of genotypes that could be compatible with additive fitness in four locus systems. We used our results for analyzing malaria drug resistance.

OUR FACULTY AND THEIR RESEARCH

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

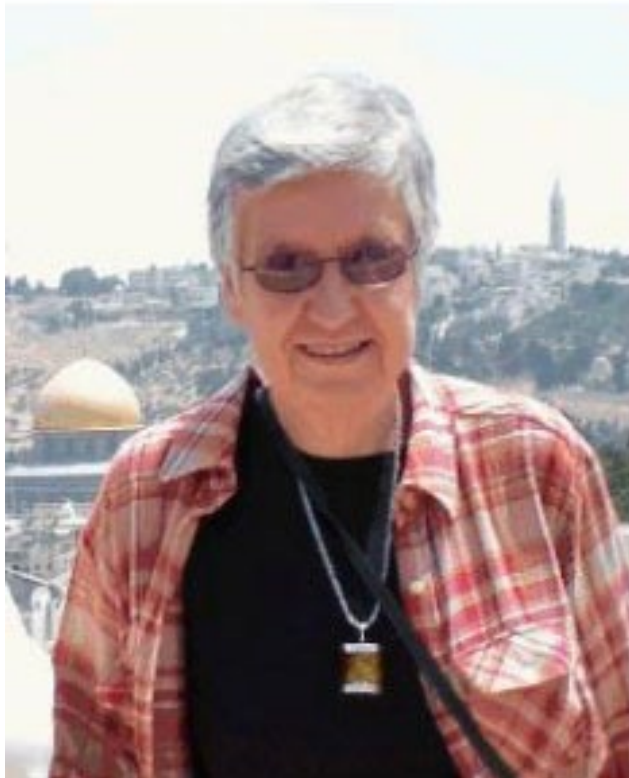


AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Mary Gray

*Applications of
statistics to human
rights, economic
equity, and education*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Monica Jackson

*Spatial statistics focusing
on detecting global
clustering patterns and
outliers for disease
surveillance.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



*Dr. Zoís
Boukounvalas*

Interpretable machine learning models and algorithms for the analysis of big multi-modal data, by combining aspects from information geometry, mathematical statistics, and numerical optimization.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



*Dr. María
Barouti*

*Data mining by identifying
patterns in big data sets
as well as developing
feature selection
algorithms*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

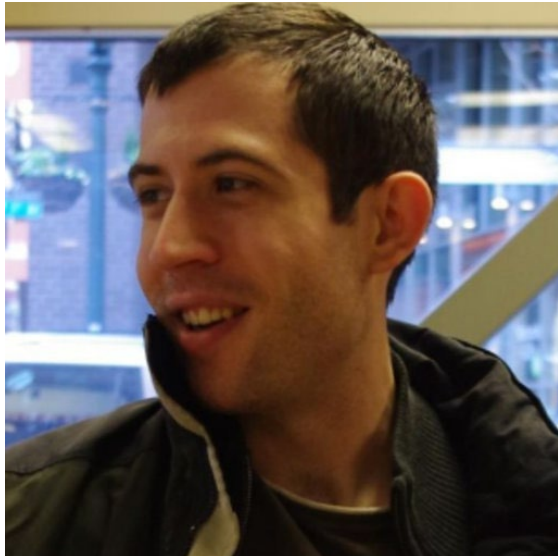


AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. David Gerard

Statistical genetics.

Bayesian analysis of the next-generation sequencing data.

Application of tensor methods to biological data

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

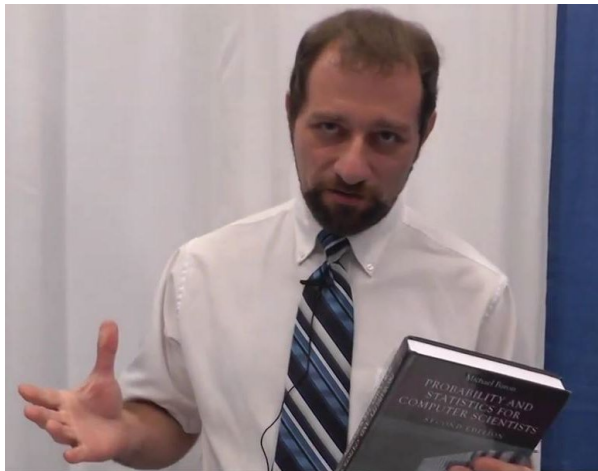


AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Michael Baron

*Sequential Analysis;
Change-Point Detection;
Bayesian inference; Multiple
Comparisons; Applications in
Clinical Trials, Epidemiology,
Insurance, and Finance.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



*Dr. Alexandra
Kapatou*

*Quality and process control;
design of experiments with
applications to clinical trials;
risk analysis, especially as
applied to food safety.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Jun Lu

*Bayesian methods and applications,
interdisciplinary statistical applications, consulting,
and statistical education.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Betty Malloy

*Functional data analysis,
biostatistics, and semiparametric
regression, with applications in
environmental statistics,
occupational health, psychology,
and other areas.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. John Nolan

*Probability theory, heavy tailed
stable and extreme value
distributions.*

*Analysis of these models is based on
mathematical theory, computational
geometry, and numerical methods*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet Our Faculty!



Julia Chifman

Systems biology and its application to molecular networks, estimation of species-level phylogenies using novel mathematical techniques, and application of methods from evolutionary biology to cancer evolution.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet Our Faculty!



Richard Ressler

*Responsible data science
and the application of
ethical approaches in data
science.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet Our Faculty!



Michael Robinson

*Signal processing, dynamics,
and applications of topology.*

*Radio propagation and
network planning, bistatic
radar processing, and
advanced radar simulation.*

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu



AMERICAN UNIVERSITY

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DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet Our Faculty!



Ahmad Mousavi

Sparse Optimization, Machine Learning, Large Scale Optimization, Mathematical Modeling, and Numerical Linear Algebra

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*



AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Jeff Gill

Gill's research interests are numerous – ranging from political behavior and institutions to statistical computing, Bayesian inference, blood/circulation, cancer epidemiology, and pediatric traumatic brain injury.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

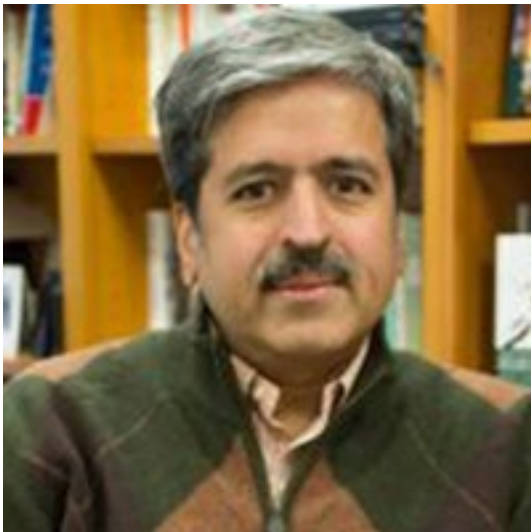


AMERICAN UNIVERSITY

WASHINGTON, D C

DEPARTMENT OF MATHEMATICS AND STATISTICS

Meet the Statistics Research Faculty!



Dr. Nímaí Mehta

As an economist and statistician, Professor Mehta's area of research is on the quality and use of data under varying institutional contexts. He is the lead coordinator of a multidisciplinary effort at the university to employ data-science, economics, and machine-learning to improve data on missing and exploited children.

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: baron@american.edu

EMPLOYMENT OPPORTUNITIES

<https://www.american.edu/cas/mathstat/statistics-ms>

Contact: *baron@american.edu*

EMPLOYMENT OPPORTUNITIES

- What do the statisticians do?

Statisticians collect, analyze, and interpret data

- Who needs their work?

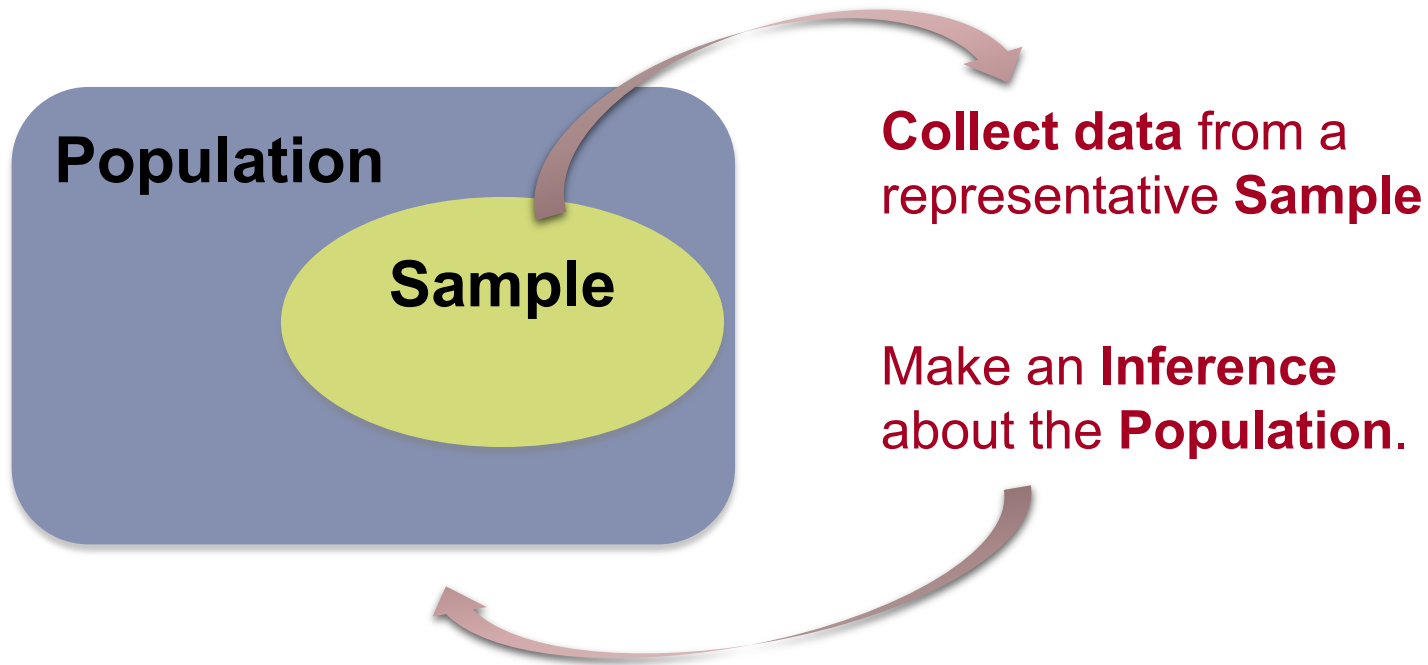
“The world is becoming quantitative. More and more professions, from the everyday to the exotic, depend on data and numerical reasoning.”

American Statistical Association

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Working as a statistician is always making discoveries



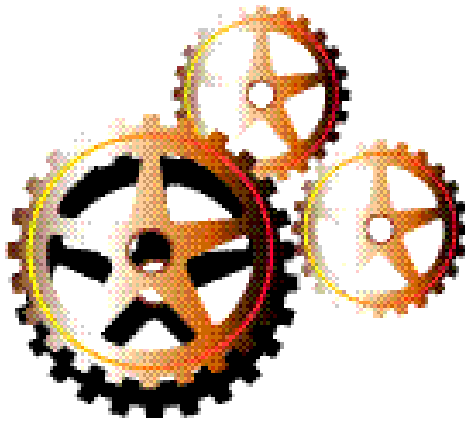
- **Collect data**
- **Turn data into useful information**
- **Make conclusions about the whole population**



Business and Industry

- **Manufacturing**

- Test quality, predict yield, determine root causes and their effect
- Identify factors that impact customer satisfaction and the company's profit



Business and Industry

- **Marketing**

- Design experiments for new products, conduct sample surveys, and perform field experiments to determine product viability



Business and Industry

- **Engineering**

- Make a consistent product, detect problems, minimize waste, and predict product life in electronics, chemicals, aerospace, pollution control, construction, and other industries



Business and Industry

Statistical Computing

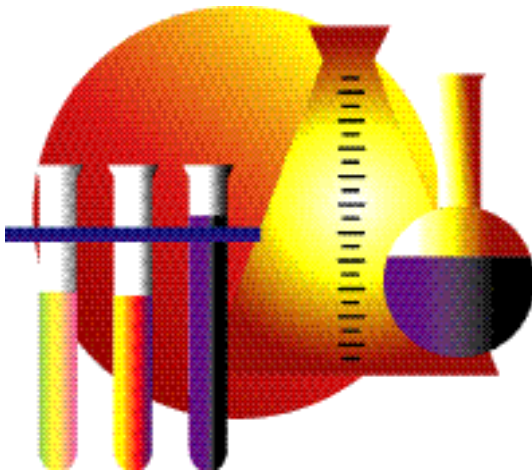
- Work in software design and development, testing, quality assurance, technical support, education, marketing, and sales to develop code that is both user-friendly and sufficiently complex



Health and Medicine

- **Epidemiology**

- Work on calculating disease incidence rates, monitor outbreaks, and detect changes in health-related behaviors such as smoking and physical activity



Health and Medicine

- **Public Health**

- Prevent disease, prolong life, and promote health through organized community efforts, including sanitation, hygiene education, diagnoses, and preventative treatment



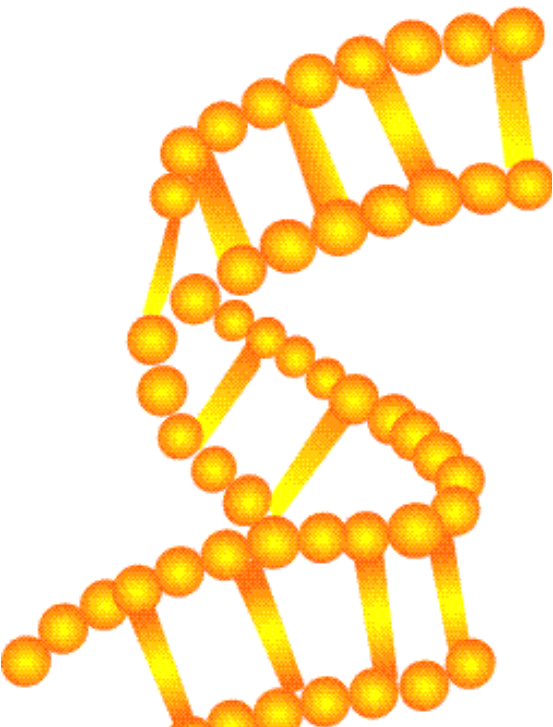
Health and Medicine

- **Pharmacology**

- Work in drug discovery, development, approval, and marketing, to ensure the validity and accuracy of findings at all stages of the process



Health and Medicine



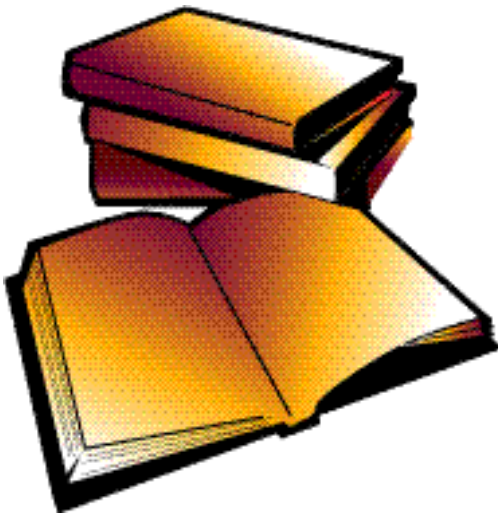
- **Genetics**

- Label possible indicators of genetic abnormalities, such as birth defects and early aging, or breed desirable characteristics in plant offspring

Learning

- **Science Writing & Journalism**

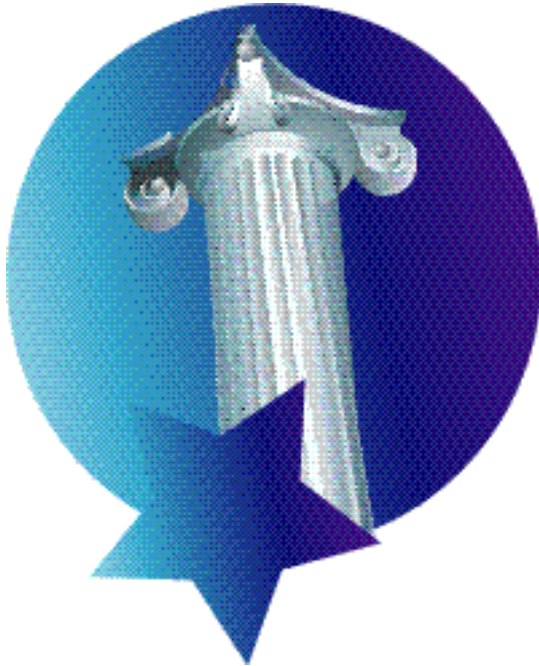
- Work with mass media, universities, and corporations to produce news briefs, articles, news releases, and other reports



Research

- **Government**

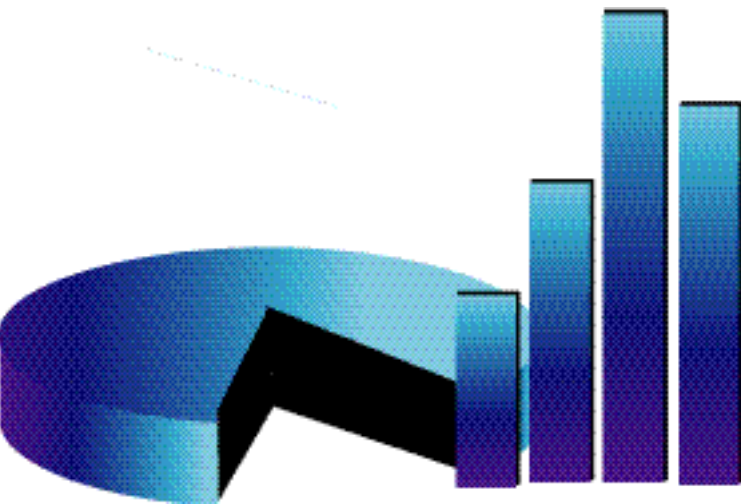
- Work in regulations for stock trading, pollution, and drug approvals, or testify in court proceedings, congressional hearings, and lobbying arguments



Research

- **Survey Methods**

- Collect data in the social sciences, education, law, forestry, agriculture, biology, medicine, business, and e-commerce, and for the government



Social Statistics

- **Law**

- Analyze data in court cases, including DNA evidence, salary discrepancies, discrimination law suits, and disease clusters



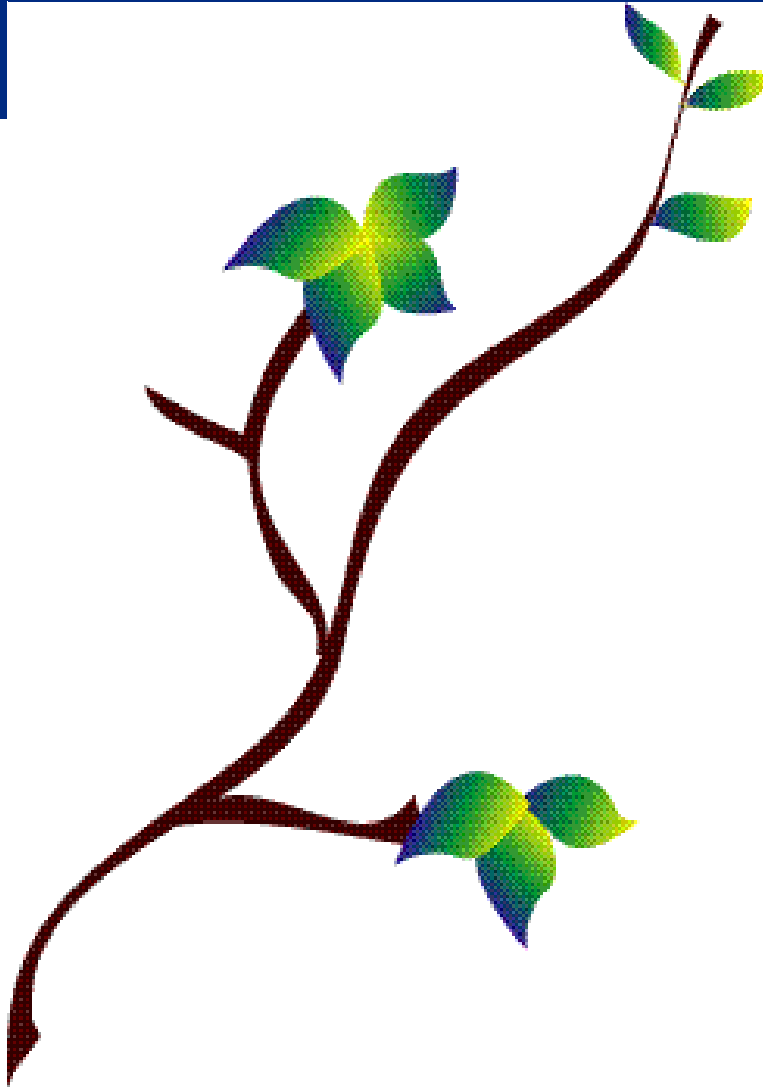
Social Statistics

- **Consulting**

- Work on a temporary basis on a variety of projects including quality improvement, pharmaceuticals, ecology, and engineering



Natural Resources



Agriculture

- Study chemical pesticides, hydrogeology, veterinary sciences, genetics, and crop management in order to ensure optimal yield

Natural Resources



- **Ecology**
 - Address questions about the earth's natural environment, including animal populations, agricultural protections, and fertilizer and pesticide safety

Opportunities around Washington DC



- The federal government employs 20% of the approximately 19,000 statisticians in the United States.
- Data collected and analyzed by statisticians in the federal government are used to make important policy decisions.
- Statisticians are employed in nearly every federal agency, offering a large variety of opportunities.
- The federal government employs statisticians with bachelor's, master's and Ph.D. degrees.



U.S. Food and Drug Administration

- Non-laboratory setting
 - Review and evaluate new drug applications
- Research setting
 - Design and analyze complex short term and long term studies

Bureau of Labor Statistics



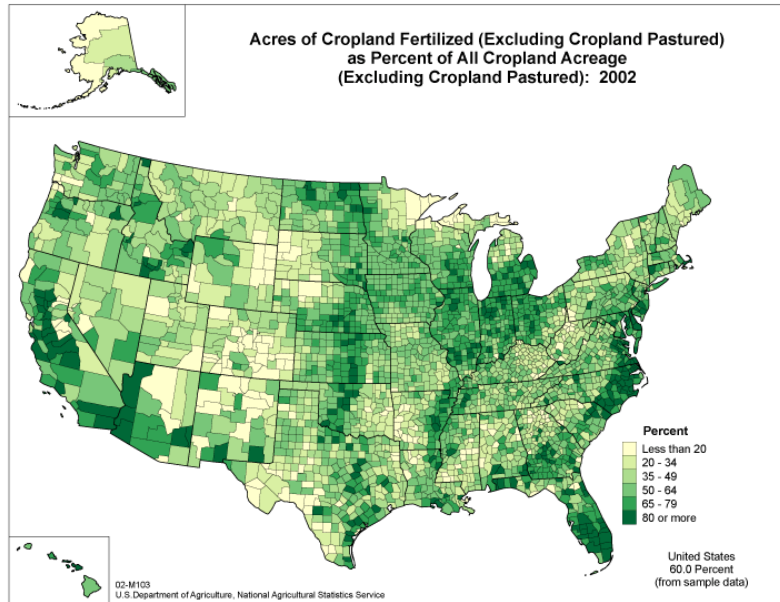
- Produce important economic indicators such as the Unemployment Rate and the Consumer Price Index



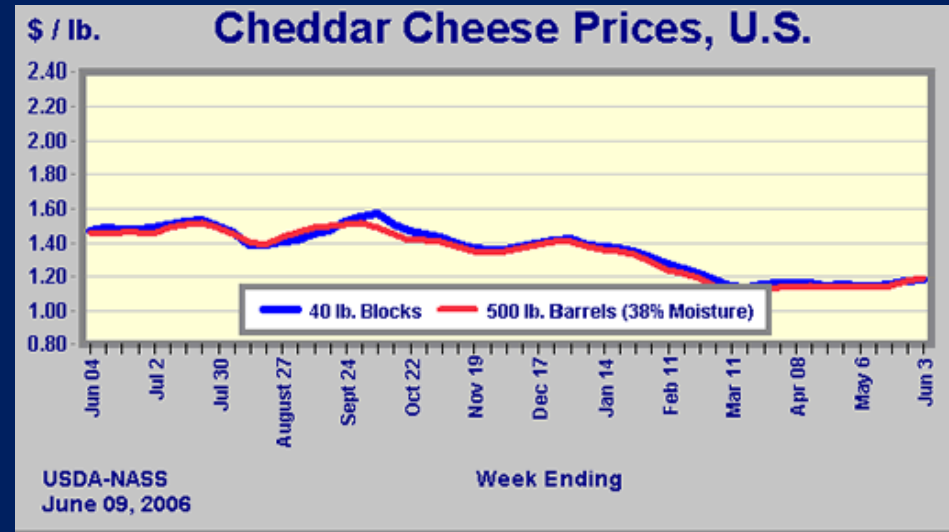
U.S. Department of Agriculture

- Agricultural Statisticians
 - Produce seasonal estimates of crops
 - Use survey statistics, mathematical statistics, and data processing to prepare data relating to many areas of agriculture
- Mathematical Statisticians
 - Research and evaluate the theory and statistical programs used in studies applied to various agricultural survey activities

What statistics does the USDA Produce?



Maps detailing the
acres of farmland
fertilized



Charts showing the price
of cheddar cheese

U.S. Census Bureau

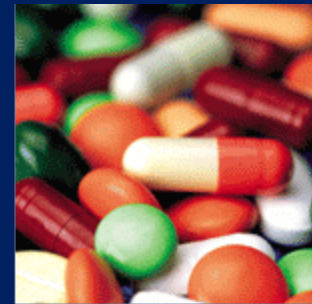


- **Statistician (Concentration on Social Science)**
 - Design, implement, analyze and report on surveys of important population trends
- **Statistician (Concentration on Economics)**
 - Collect information from retailers on the amount of retail trade
- **Mathematical Statistician**
 - Develop new statistical methodologies and perform innovative research to solve statistical problems



National Institutes of Health

Design clinical trials,
analyze large datasets
coming from genetics,
DNA sequencing, MRI,
fMRI, and other
medical diagnostics
tools, etc.





Centers for Disease Control and Prevention



CDC employs statisticians to design both laboratory and epidemiologic studies and analyze the data from individual projects.

CONTACT INFORMATION

AMERICAN UNIVERSITY MATHEMATICS & STATISTICS DEPARTMENT

<https://www.american.edu/cas/mathstat/statistics-ms/>

PROGRAM DIRECTOR, MS in STATISTICS

Dr. Michael Baron, baron@american.edu

WEEKLY Ask-Me-Anything ZOOM SESSIONS ABOUT MS in STATISTICS:

<https://american.zoom.us/j/95868480371> - every Monday at 12 noon

APPLY to the PROGRAM:

<https://www.american.edu/cas/admissions/apply.cfm>